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II Semester M.C.A. Examination, October/November 2020
(CBCS Scheme)
COMPUTER SCIENCE
MCA 204T : Operating Systems

Time : 3 Hours Max. Marks : 70

Instructions : *1) Part – **A** : Answer **any five** questions.*
*2) Part – **B** : Answer **any four** questions.*

PART – A

Answer **any five** questions. **Each** question carries **six** marks. **(5×6=30)**

1. What is a process ? Explain different states of a process with neat diagram.
2. Discuss various methods used to implement inter-process communication among cooperating processes with suitable examples.
3. What is critical section ? Illustrate a software-based Peterson's solution to the critical section problem.
4. Elaborate on 3 different steps in binding of instructions and data to memory addresses with necessary diagrams.
5. Briefly explain any two page replacement algorithms with examples.
6. What are the causes for thrashing ? List the methods used to prevent thrashing.
7. Distinguish between single-level and two-level directory with supporting diagrams.
8. Explain security mechanisms used in Linux system.

PART – B

Answer **any four** questions. **Each** question carries **10** marks. **(4×10=40)**

9. a) Explain FCFS and Round Robin Scheduling algorithms with example. **6**
   b) Write a short note on Multilevel Feedback-Queue scheduling. **4**

10. a) Define semaphore. Explain the implementation of semaphore in solving Dining-Philosophers problem. **6**
    b) Explain Readers-Writers Problem in process synchronization. **4**

P.T.O.

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11. a) Draw the Gantt chart and calculate average turnaround time using FCFS and non-preemptive SJF Scheduling. 6

Process	Arrival Time	Burst Time in ms
P_1	0.0	6
P_2	0.0	4
P_3	0.0	1
P_4	0.0	5

b) Explain different types of schedulers. 4

12. a) Using FIFO and LRU page replacement algorithms, do a page trace analysis with three page frames for the following reference string : 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5. Compute page faults for both the algorithms. 6

b) How memory reference is made to the page table with Translation look-aside buffer ? Explain with a diagram. 4

13. Explain different types of disk scheduling algorithms. 10

14. Write short notes on :

a) Context-switching. 4

b) Compaction. 3

c) Attributes in a File. 3

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PS – 386

II Semester M.C.A. Examination, November/December 2022
(CBCS) (2020-21 and Onwards)
COMPUTER SCIENCE
2MCA 1 : Operating System

Time : 3 Hours Max. Marks : 70

Instruction : *Answer **any five** questions from Section – **A** and **any four** questions from Section – **B**.*

SECTION – A

I. Answer **any five** of the following questions. **Each** carries **6** marks. **(5×6=30)**

1) Explain the different operating system structures.
2) Define system call. Describe the system call related to process management.
3) What is critical section problem ? Explain solution to consumer/producer problem.
4) Write difference between paging and segmentation.
5) What is disk scheduling ? Explain FCFS and SCAN disk scheduling.
6) How can deadlocks be prevented ? Describe them.
7) What is thrashing ? How do we overcome thrashing ?
8) By using access matrix, how do we secure an operating system ?

SECTION – B

II. Answer **any four** questions. **Each** carries **10** marks. **(4×10=40)**

9) a) In what ways two process communicate with each other ? **4**
   b) Explain three multithreaded models. **6**
10) a) Explain Peterson's solution to the mutual exclusion problem. **7**
   b) What is critical section ? Given a variable x and two process wish to write on that variable, at the same time. How does OS manage such a situation ? **3**

P.T.O.

11) a) What is a page table ? What are the various mechanisms for realising page table ? 5

b) Discuss contiguous memory allocation. 5

12) a) Explain FIFO page replacement algorithm with an example. 5

b) Explain allocation of frames methods. 5

13) a) Write any 5 difference between Windows and Linux Operating System. 5

b) Explain types of VMs and their implementations. 5

14) Write short notes on : 10

i) Storage Management

ii) Semaphores.

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II Semester M.C.A. Examination, November/December 2023
(CBCS) (2020 – 21 and Onwards)
COMPUTER APPLICATIONS
2MCA1 : Operating Systems

Time : 3 Hours Max. Marks : 70

Instruction *: Answer* ***any five*** *from Part –* ***A*** *and* ***any four*** *from Part –* ***B****.*

PART – A

Answer **any five** of the following. **Each** question carries **six** marks. **(5×6=30)**

1. Define system call. How does a system call interface with the Kernel of an operating system ?
2. Explain the process state with a state transition diagram.
3. Discuss the solution to the 2-process critical section problem. Mention the issues in this solution.
4. Explain Readers-Writers problem of synchronization.
5. What is a resource allocation graph ? How can you ascertain the presence of a deadlock from a graph ?
6. Write the difference between paging and segmentation.
7. Describe the shortest seek time first and C-Scan disk scheduling algorithm.
8. Narrate the distributed system and the advantages of distributed system.

PART – B

Answer **any four** of the following. **Each** question carries **ten** marks. **(4×10=40)**

9. Discuss the concept of computer system organization and its significance in the design and functionality of computing systems. **10**
10. a) Suggest and explain one method each to avoid "Hold and Wait" and "Circular Wait" conditions for deadlock prevention. **5**
    b) Define semaphores. Explain the implementation of semaphore in solving dining-philosophers problem. **5**

P.T.O.

11. Considering a system with five processes P0 through P4 and three resources of type A, B, C. Resource type A has 10 instances, B has 5 instances and type C has 7 instances. Suppose at time t0 following snapshot of the system has been taken : 10

Process	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	7	5	3	3	3	2
P1	2	0	0	3	2	2			
P2	3	0	2	9	0	2			
P3	2	1	1	2	2	2			
P4	0	0	2	4	3	3			

Answer the following questions :

a) What will be the content of the need matrix ?

b) Is the system in a safe state ? If yes, then what is the safe sequence ?

c) What will happen if Process P1 requests one additional instance of resource type A and two instances of resource type C ?

12. A) What is a file ? Explain the various file allocation methods. 7

B) Write short notes on a virtual machine. 3

13. A) Explain FCFS and SJF scheduling algorithms. 6

B) Write a short notes on :

i) Starvation and aging. 2

ii) Swapping. 2

14. Explain the concept of an access matrix in computer security. Provide an example scenario where an access matrix would be applicable and describe how it helps in managing access control in that scenario. 10

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II Semester M.C.A. Examination, December 2024
(CBCS) (2020-21 and Onwards)
COMPUTER SCIENCE
2MCA1 : Operating Systems

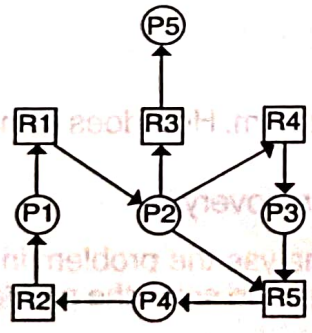
Time : 3 Hours Max. Marks : 70

***Instruction** : Answer **all** the Sections.*

SECTION – A

Answer **any five** of the following. **Each** question carries **six** marks. **(5×6=30)**

1. "Operating system typically utilize the Kernel mode and user mode functions as rudimentary form of protection". Explain the two modes and justify the statement.
2. Distinguish between process and thread outline the different multithreading models with neat sketches.
3. What are semaphores and list their advantages ? Also explain primitive semaphore operations.
4. Consider the following page reference.
   String : 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 7 0 1
   For a memory with three frames, how many page faults would occur for LRU and FIFO page replacement algorithm ? Which is most efficient among them.
5. Discuss various directory implementation methods.
6. How do you detect a deadlock in Resource Allocation Graph (RAG) consisting of exactly 1 instance of each resource type ? Draw a corresponding wait for graph for the RAG shown below and determine whether a deadlock is detected.



7. What is Protection ? Discuss principles for protection.
8. State and explain the FCFS, SSTF and SCAN disk scheduling with examples.

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SECTION – B

Answer **any four** of the following. **Each** question carries **10** marks. **(4×10=40)**

9. a) List out various services provided by the OS that are helpful to the users. **5**

   b) Describe the "Layered Approach" to the design of an operating system and state its advantages. **5**

10. Consider the following data about processes. **10**

Process	Arrival time	Burst time
P1	0	7
P2	3	2
P3	4	3
P4	4	1
P5	5	3

   i) Draw charts to illustrate execution using FCFS, preemptive SRTF and RR (TS = 2 ms).

   ii) Compute waiting time and turn around time for each process.

11. Consider the following system snapshot using data structures in Banker's Algorithm, with resources A, B, C, D and process P0 to P4. **10**

	Max				Allocation				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	6	0	1	2	4	0	0	1	3	2	1	1
P1	1	7	5	0	1	1	0	0				
P2	2	3	5	6	1	2	5	4				
P3	1	6	5	3	0	6	3	3				
P4	1	6	5	6	0	2	1	2				

   a) Compute need matrix.

   b) Is the system in a safe state ? Why ?

12. a) Discuss Peterson's solution to critical section problem. How does it ensure mutual exclusion. **5**

   b) Discuss various approaches used for deadlock recovery. **5**

13. With neat diagram explain paging technique and analyse the problem in simple paging technique. Also explain how TLB is used to solve the problem. **10**

14. Write short note on :

   a) Linux Kernel. **3**

   b) Virtualization. **3**

   c) Thrashing. **4**

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II Semester M.C.A. Examination, November/December 2025
(CBCS) (2020 – 21 and Onwards)
COMPUTER SCIENCE
2MCA1 : Operating Systems

Time : 3 Hours Max. Marks : 70

Instruction : *Answer **all** Sections.*

SECTION – A

Answer **any five** of the following. **Each** question carries **six** marks. **(5×6=30)**

1. One of the most important aspects of operating systems is to multiprogram. Discuss the idea behind, as to how OS can provide an environment to realize this.

2. Consider a scenario where a process is currently in "Running" state. Discuss the various events which can cause the process to transition to "Ready" state in detail with an appropriate diagram.

3. Develop a pseudocode to implement mutual exclusion for the Dinning Philosopher problem.

4. What is deadlock ? What are the necessary conditions for the deadlock to occur ?

5. Given the memory partitions of 200 K, 700 K, 500 K, 300 K, 100 K, 400 K. Apply first fit and best fit to place 315 K, 427 K, 250 K and 550 K.

6. Write a note on paging and swapping.

7. Given the following sequences :

   95, 180, 34, 119, 11, 123, 62, 64 with the head initially at track 50 and ending at track 199. What is the total disk movement by the disk arm to satisfy the I/O request using FCFS, SSTF and SCAN algorithm ?

8. What is "need-to-know" principle ? Why it is important for the protection system to adhere to this principle ?

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SECTION – B

Answer **any 4** of the following. **Each** question carries **10** marks. **(4×10=40**

9. Consider the following example wherein content of the file 'src' has to be copied to file 'dost'. Discuss what are the various system calls used in the process. Also categorize system calls into different types based on their functionality.

10. Consider the following arrival time (ms) and burst time (ms). Draw Gantt chart and calculate waiting time and turn around time using preemptive and non-preemptive priority scheduling. Discuss which is optimum.

Process	Arrival time	Burst time	Priority
P1	0	6	4
P2	3	5	2
P3	3	3	6
P4	5	5	3

Note : Larger priority has higher value.

11. Consider the following snapshot of the system.

	Allocation				MAX				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	0	1	2	0	0	1	2	1	5	2	0
P1	1	0	0	0	1	7	5	0				
P2	1	3	5	4	2	3	5	6				
P3	0	6	3	2	0	6	5	2				
P4	0	0	1	4	0	6	5	6				

Using Banker's Algorithm, answer the following :

i) What is the content of the matrix 'NEED' ?

ii) Is the system in safe state ? If yes, give the safe state.

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12. Given a system with 3 processes P1, P2, P3 and 3 resources R1, R2, R3 (1 instance each).

    * P1 holds R1 and requests R2

    * P2 holds R2 and requests R3

    * P3 holds R3 and requests R1

    a) Draw the resource allocation graph. 4

    b) Convert it to wait for graph. 3

    c) Detect if the system is in a deadlock. Justify your answer. 3

13. Use FIFO page replacement algorithm for the given reference string for a memory with two, three and four frames 1, 2, 3, 5, 2, 3, 5, 7, 2, 1, 2, 3, 8, 6, 4, 3, 2, 2, 3, 6. And determine whether it suffers from Belady's Anomaly ? All page frames are empty, so that first unique pages will all cost one fault each. 5

14. a) Explain access matrix method of system protection with domain as objects. 5

    b) Write short note on virtual machine.

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